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In this hands-on workshop you will build an agentic AI application from scratch using Python and the AWS SDK — no frameworks, no abstractions. In about an hour you will have a working weather agent and, more importantly, a clear mental model of how agents are put together.

What You'll Build

A weather agent that takes a location in plain English (a city name, a ZIP code, or a vague description like "national park near Homestead in Florida"), figures out the right National Weather Service API call, runs it, and returns a readable forecast.

The agent is built from four functions, each with a single responsibility. We introduce them in section 1 and write them in section 4.

Workshop Structure

1. **What is Agentic AI?** (10 min) — The Problem → Plan → Action → Result loop and the four functions that implement it.
2. **Environment Setup** (12 min) — Python, AWS SDK, and Bedrock model access.
3. **Understanding the Architecture** (8 min) — The six-step workflow and how the pieces fit.
4. **Building the CLI Agent** (23 min) — Writing each of the four functions and the main loop.
5. **Running the Web App** (5 min) — The same agent behind a Streamlit interface.
6. **Key Takeaways** (2 min) — Patterns to reuse on your own agents.
7. **Clean Up** (2 min) — Tear down anything you provisioned.

Prerequisites

- No prior AI/ML experience required
- Basic Python (variables, functions, if-statements)
- An AWS account, or access to a workshop environment provided by AWS

Learning Objectives

By the end you will:

- Understand what makes a system "agentic" without relying on framework jargon
- Build an AI agent in pure Python with `boto3`
- Call external APIs based on AI-generated plans
- Process raw API output back into a useful answer
- Recognise the same pattern in other domains so you can apply it to your own work

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